22

Lecture 22

Quality of Service (QoS) in 5G

کیفیت خدمات (QoS) در نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 7, 9, 13, 20 and 21; M/M/1 queue

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-22-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the 5G QoS model: QoS flows, QFI, QoS profiles, QoS rules and their mapping to radio bearers.
- **LO2** **Interpret** standardised 5QI characteristics (resource type, priority, packet delay budget, packet error rate, maximum data burst volume).
- **LO3** **Trace** how policy from the PCF becomes QoS flows enforced in UE, RAN and UPF.
- **LO4** **Compute** delay budgets, HARQ needs for a packet-error target and the effect of priority scheduling.
- **LO5** **Compare** the 5G flow-based model with LTE's bearer-based model.

2 Prerequisite knowledge

You need LTE scheduling and HARQ (Lecture 7), the EPS bearer and QCI model (Lecture 9), PFCP rules (Lecture 13), PDU session establishment (Lecture 20) and slicing (Lecture 21). The priority model extends the M/M/1 queue of Lecture 12.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
QoS flow	جریان کیفیت خدمات	Finest granularity of QoS differentiation in a PDU session.
QFI	شناسهٔ جریان QoS	6-bit identifier of a QoS flow, carried in N3 encapsulation.
5QI	شاخص QoS نسل پنجم	Pointer to a set of standardised QoS characteristics.
QoS profile	نمایهٔ QoS	Parameters given to the RAN per QoS flow (5QI, ARP, GFBR, MFBR, ...).
QoS rule	قاعدهٔ QoS	UE-side packet filters mapping uplink traffic to QoS flows.
GBR / non-GBR	با / بدون نرخ تضمین‌شده	Resource types; delay-critical GBR for URLLC.
PDB / PER	بودجهٔ تأخیر / نرخ خطای بسته	Upper bounds on delay and loss for a 5QI.
MDBV	بیشینهٔ حجم انفجار داده	Largest burst to be served within the PDB (delay-critical GBR).
SDAP	پروتکل سازگاری دادهٔ خدمت	RAN layer mapping QoS flows to data radio bearers.
Reflective QoS	بازتابی QoS	UE derives uplink QoS rules from downlink packets.

3 Introduction and motivation

Different applications need different treatment: a voice call needs low delay and tolerates some loss; a file transfer needs no loss but tolerates delay; a robot controller needs both low delay and extremely low loss; a background update needs neither. A mobile network cannot satisfy all of these with a single best-effort pipe, because the scarce resource – radio capacity – must be shared under load. QoS mechanisms decide which packets go first, which are guaranteed a rate, and which may be dropped.

LTE solved this with *bearers*: one per QoS class, each requiring signalling to set up along the whole path. 5G replaces bearers in the core with lightweight *QoS flows*, marked per packet and mapped flexibly to radio bearers by the RAN [5]. The resulting model is more scalable, supports URLLC with new delay-critical characteristics, and integrates with slicing (Lecture 21): a slice chooses the network, QoS flows the treatment inside it.

4 Fundamental concepts

4.1 QoS flows

A PDU session contains one or more QoS flows, each identified by a QFI (Figure 1). Every PDU session has a default QoS flow (non-GBR) that carries traffic not matching any other rule. Classification happens at the edges:

- **downlink**: the UPF classifies packets with PDRs (packet filters from PCC rules) and marks them with the QFI in the N3 encapsulation header;
- **uplink**: the UE classifies packets with *QoS rules* (packet filters with precedence) and the SDAP layer marks them;
- **RAN**: the gNB maps QoS flows to data radio bearers (DRBs) as it sees fit – several flows may share a DRB.

Because QFI marking is per packet, adding a QoS flow does not require a new tunnel: all flows of a PDU session share one N3 tunnel.

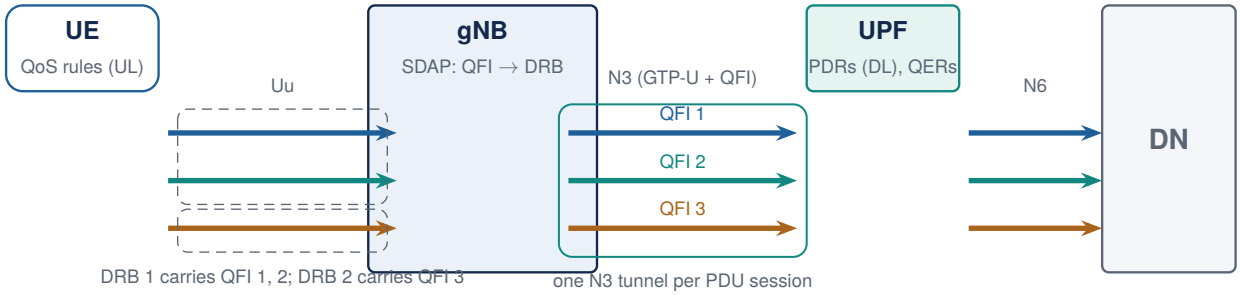


Figure 1. 5G QoS model: classification in UE and UPF, QFI marking on N3, flow-to-DRB mapping by SDAP in the gNB.

4.2 QoS parameters

Definition 1 • QoS profile and characteristics

The SMF gives the RAN a *QoS profile* per QoS flow: the 5QI, the allocation and retention priority (ARP), and for GBR flows the guaranteed and maximum flow bit rates (GFBR, MFBR), optionally notification control and a maximum packet loss rate. The 5QI points to *QoS characteristics*: resource type (non-GBR, GBR, delay-critical GBR), priority level, packet delay budget (PDB), packet error rate (PER), averaging window and, for delay-critical GBR, the maximum data burst volume (MDBV) [5].

Aggregate limits apply to non-GBR traffic: the *Session-AMBR* per PDU session (enforced by UPF and UE) and the *UE-AMBR* across a UE's sessions (enforced by the RAN).

4.3 Packet delay budget and packet error rate

The PDB is an upper bound on the delay between UE and the UPF's N6 termination, for a given confidence; it includes a core-network part and an access-network part. The PER is an upper bound on the rate of packets processed by the link layer but not delivered to the upper layer. For GBR flows, packets delayed beyond the PDB count as lost (for delay-critical GBR, they are counted in the PER), so PDB and PER together define the reliability target of a URLLC flow.

Table 1. Selected standardised 5QI values (TS 23.501, Table 5.7.4-1) [5]. Lower priority values mean higher priority.

5QI	Resource type	Priority	PDB	PER	Example services
1	GBR	20	100 ms	10^{-2}	Conversational voice
2	GBR	40	150 ms	10^{-3}	Conversational video (live)
3	GBR	30	50 ms	10^{-3}	Real-time gaming, V2X messages
4	GBR	50	300 ms	10^{-6}	Non-conversational video (buffered)
65	GBR	7	75 ms	10^{-2}	Mission-critical push-to-talk voice
5	Non-GBR	10	100 ms	10^{-6}	IMS signalling
6	Non-GBR	60	300 ms	10^{-6}	Buffered video, TCP applications
7	Non-GBR	70	100 ms	10^{-3}	Voice, live video, interactive gaming
9	Non-GBR	90	300 ms	10^{-6}	Default bearer traffic (web, e-mail)
82	Delay-critical GBR	19	10 ms	10^{-4}	Discrete automation (MDBV 255 bytes)
85	Delay-critical GBR	21	5 ms	10^{-5}	Electricity distribution, high voltage (MDBV 255 bytes)

5 Detailed technical explanation

5.1 From policy to QoS flows

QoS starts with policy (Figure 2). An application function (e.g. the IMS for a voice call) asks the PCF for treatment of a media flow; the PCF produces PCC rules with 5QI, ARP and bit rates and sends them to the SMF. The SMF binds each PCC rule to a QoS flow – an existing one with the same parameters or a new one with a new QFI – and installs: PDRs and QERs in the UPF (N4), the QoS profile in the gNB (N2), and the QoS rule in the UE (NAS). The gNB performs admission control for GBR flows and configures DRBs and the SDAP mapping.

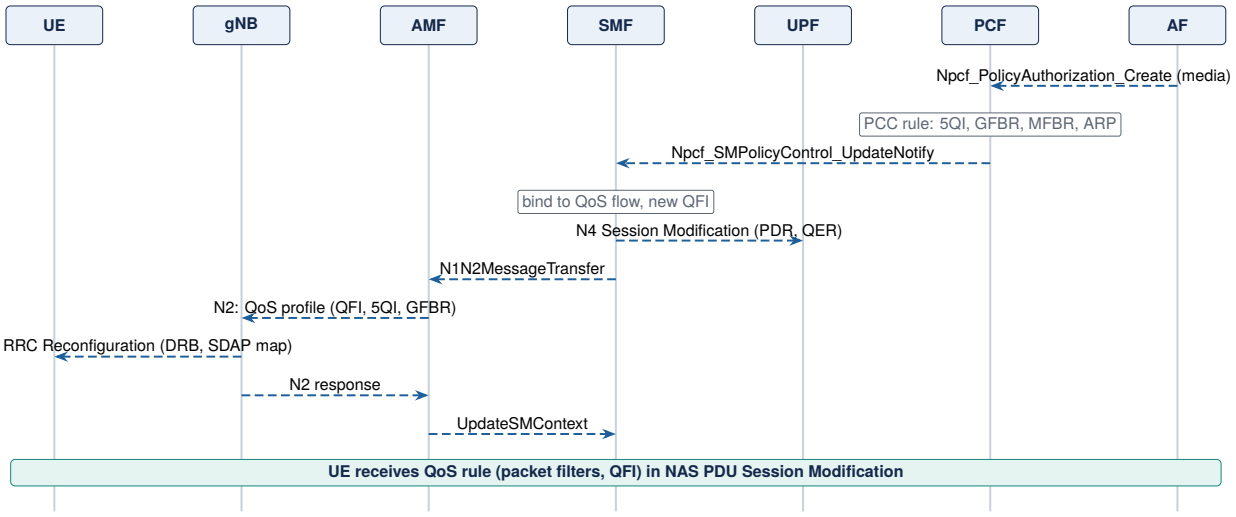


Figure 2. Establishing a GBR QoS flow triggered by an application function (simplified).

5.2 Reflective QoS

For applications with many short flows (e.g. web browsing with different priorities) signalling a QoS rule for each would be costly. With *reflective QoS* the UPF sets a reflective QoS indicator (RQI) in downlink packets; the UE then creates an uplink QoS rule mirroring the downlink packet's addresses and ports, with the same QFI, without NAS signalling.

5.3 Enforcement in the RAN

The gNB scheduler translates QoS profiles into scheduling decisions: priority levels order the queues, GFBR is guaranteed over the averaging window, PDB determines how long a packet may wait (and which numerology, mini-slots or configured grants are needed), and PER determines the target BLER and HARQ/repetition strategy. QoS notification control lets the gNB inform the SMF when a GFBR can no longer be guaranteed, so that the application can adapt (e.g. lower video resolution).

5.4 QoS monitoring

Since Release 16 the network can measure packet delay per QoS flow between UE, gNB and UPF and report it to the SMF and, via the NEF, to application functions – the basis for SLAs on URLLC slices and for applications that adapt to measured latency.

6 Mathematical models

6.1 Splitting the delay budget

Model. The PDB is shared between the core network (UPF to gNB, D_{CN}) and the access network (D_{AN}):

$$D_{\mathrm{AN}} = \mathrm{PDB} - D_{\mathrm{CN}}. \tag{1}$$

Example. 5QI 1 (100 ms) with an assumed core part of 20 ms leaves 80 ms for scheduling, HARQ and transmission – comfortable. 5QI 85 (5 ms) with a UPF 50 km away (0.25 ms one way in fibre plus processing, say 1 ms) leaves 4 ms: the UPF must be at the edge and the radio must use short slots.

6.2 Rate needed for a data burst

Model. For a delay-critical GBR flow, a burst of up to MDBV bytes must be delivered within the access-network budget. The instantaneous rate the scheduler must provide during the burst is

$$R_{\mathrm{burst}} \geq \frac{8\,\mathrm{MDBV}}{D_{\mathrm{AN}}}. \tag{2}$$

Example. 5QI 82 (MDBV 255 bytes, PDB 10 ms) with $D_{\mathrm{AN}} = 9$ ms: 227 kbit/s; 5QI 85 (255 bytes, PDB 5 ms) with $D_{\mathrm{AN}} = 4$ ms: 510 kbit/s. The rates are small; the challenge is delivering them *reliably* within milliseconds.

6.3 HARQ transmissions for a PER target

Model. If each transmission of a block fails independently with probability p (the target BLER), the residual error after n transmissions is p^n. To reach a target PER,

$$n \geq \frac{\ln \mathrm{PER}}{\ln p}, \tag{3}$$

and all n transmissions must fit into D_{AN}.

Example. PER $= 10^{-5}$: with $p = 0.1$, $n = 5$ transmissions (Figure 3); with a conservative MCS giving $p = 0.01$, $n = 3$. If one HARQ round trip takes 4 slots of 0.125 ms (120 kHz) = 0.5 ms, five transmissions need 2–2.5 ms – feasible within 4 ms; at 15 kHz (1 ms slots) they would need about 20 ms – impossible. URLLC therefore combines robust MCS (low p), short slots and sometimes blind repetitions.

Limitation. Errors are not independent when the channel is in a deep fade; diversity (frequency, antennas, multi-connectivity) is needed to make the independence assumption approximately valid.

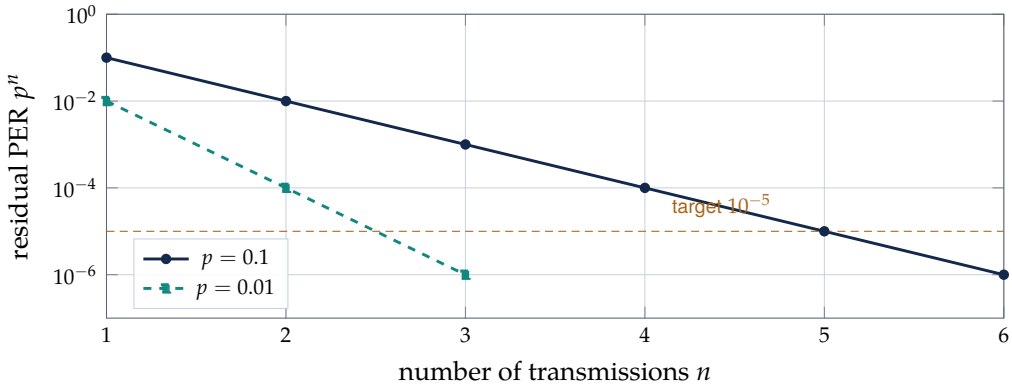


Figure 3. Residual packet error rate after n transmissions (Eq. (3)).

6.4 Priority scheduling

Model. Two classes share a link modelled as M/M/1 with service rate μ; class 1 (load ρ_1) has preemptive priority over class 2; total load ρ. The mean sojourn times are

$$T_1 = \frac{1}{\mu(1-\rho_1)}, \qquad T_2 = \frac{1}{\mu(1-\rho_1)(1-\rho)}, \tag{4}$$

compared with $T = 1/(\mu(1-\rho))$ for FIFO.

Example. $\mu = 1000$ packets/s, $\rho_1 = 0.2$: $T_1 = 1.25$ ms regardless of the total load; at $\rho = 0.9$, $T_2 = 12.5$ ms against 10 ms for FIFO (Figure 4).

Interpretation. Priority protects the high-priority class completely from the load of lower classes, at a moderate cost to the others – as long as the high-priority load itself stays small. This is why GBR and delay-critical flows need admission control: without it, too much "high-priority" traffic destroys the protection.

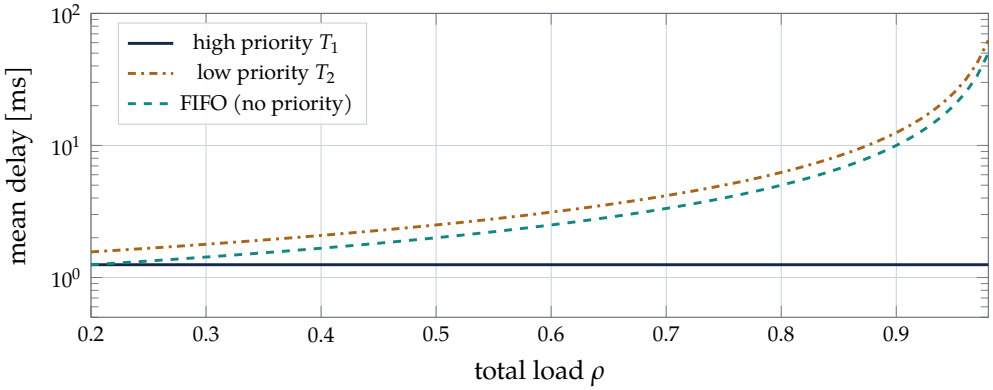


Figure 4. Mean delay with preemptive priority and with FIFO (Eq. (4), $\rho_1 = 0.2$, $\mu = 1000\,\mathrm{s}^{-1}$).

7 Network architecture and diagrams

QoS is enforced at three points: the UE (uplink classification and Session-AMBR), the gNB (scheduling per QoS flow and DRB, UE-AMBR, GFBR) and the UPF (downlink classification, gating, MFBR and Session-AMBR, usage reporting). The PCF decides; the SMF translates decisions into rules for each point (Figures 1 and 2). The transport network between gNB and UPF should map QFIs or 5QIs to DSCP classes so that it does not undo the differentiation.

8 Step-by-step procedures

Procedure 1 • Adding a GBR QoS flow for a voice call

1. The IMS (AF) sends the call's media description to the PCF (Npcf_PolicyAuthorization).
2. The PCF derives a PCC rule (5QI 1, GFBR/MFBR from the codec, ARP) and notifies the SMF.
3. The SMF allocates a QFI, installs a PDR and QER in the UPF and sends the QoS profile to the gNB and the QoS rule to the UE through the AMF.
4. The gNB admits the flow, maps it to a DRB with suitable RLC mode and scheduling, and confirms; voice packets now receive GBR treatment.
5. At call end the PCF removes the rule; the SMF releases the QoS flow in UPF, RAN and UE.

Procedure 2 • Checking a URLLC QoS design

1. From the 5QI, read PDB, PER and MDBV; subtract the core-network delay to obtain D_{AN} (Eq. (1)).
2. Compute the burst rate from MDBV (Eq. (2)) and the number of transmissions from PER and BLER (Eq. (3)).
3. Choose numerology, mini-slots or configured grants so that all transmissions fit into D_{AN}.

4 Check the priority load: the high-priority traffic must remain a small fraction of capacity (Eq. (4)); configure admission control.

9 Worked examples

Worked example 1 • Choosing 5QIs

Problem. Assign 5QIs to: VoNR voice, IMS SIP signalling, a live video call, web browsing, factory motion control.
Solution. Voice 1; SIP 5; live video 2 (or 7 if non-GBR); web 9 (default) or 6/8; motion control a delay-critical GBR value such as 82 (or 85 if 5 ms is required), with the UPF at the edge.

Worked example 2 • URLLC feasibility

Problem. A flow with 5QI 85 carries 200-byte bursts. The UPF is on site (core delay 0.5 ms). With 30 kHz SCS (0.5 ms slots), a HARQ round trip of 4 slots and $p = 0.01$, can the PER be met?
Solution. $D_{\mathrm{AN}} = 4.5$ ms; burst rate $8 \times 200/0.0045 = 356$ kbit/s; $n = \ln 10^{-5}/\ln 0.01 = 2.5 \to 3$ transmissions needing $3 \times 2 = 6$ ms – too long. Options: mini-slots or 120 kHz SCS (HARQ RTT 0.5 ms: 1.5 ms total), or blind repetitions within one slot window.

Worked example 3 • Priority protection

Problem. With Eq. (4), what happens to T_1 if the high-priority load grows from 0.2 to 0.6, and to T_2 at total load 0.9?
Solution. $T_1 = 1/(1000 \times 0.4) = 2.5$ ms (doubled); $T_2 = 1/(1000 \times 0.4 \times 0.1) = 25$ ms. High-priority traffic must itself be limited.

10 Technology comparison

Table 2. LTE bearer-based QoS versus 5G flow-based QoS.

	LTE (EPS bearers)	5G (QoS flows)
Unit of QoS	EPS bearer (radio + S1 + S5 bearer)	QoS flow, marked by QFI
Core tunnels	One per bearer	One per PDU session
Radio mapping	One radio bearer per EPS bearer	Flexible flow-to-DRB mapping (SDAP)
QoS index	QCI (1–9, 65–80)	5QI, including delay-critical GBR
Uplink classification	UL TFT, signalled	QoS rules, signalled or reflective
New flow	Dedicated bearer set-up end to end	QoS rule/profile update, no new tunnel
Delay monitoring	Limited	Per-flow QoS monitoring (Rel-16)

Same ideas, lighter mechanics. The QoS parameters (priority, PDB, PER, ARP, GBR/MBR, AMBR) are almost identical to LTE's; what changed is the mechanism – marking instead of tunnels, flexible RAN mapping, and reflective rules – which makes fine-grained QoS affordable.

11 Key takeaways

Key takeaways

- 5G QoS is flow-based: QFI marking per packet, one N3 tunnel per PDU session, SDAP maps flows to DRBs.
- 5QI gives resource type, priority, PDB, PER (and MDBV for delay-critical GBR); ARP and GFBR/MFBR complete the QoS profile.
- PCF → SMF → (UPF via N4, gNB via N2, UE via NAS): policy becomes enforceable rules.
- $D_{\text{AN}} = \text{PDB} - D_{\text{CN}}$; burst rate $\geq 8\,\text{MDBV}/D_{\text{AN}}$; transmissions $n \geq \ln \text{PER}/\ln p$.
- Priority protects high-priority traffic only if its own load is limited: $T_1 = 1/(\mu(1-\rho_1))$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Flows and bearers

A UE has an Internet PDU session with three QoS flows. How many N3 tunnels and how many DRBs does it need at least and at most?

Exercise 2 • Level 1 – conceptual • Reflective QoS

Explain why reflective QoS saves signalling and give one situation where it should not be used.

Exercise 3 • Level 2 – numerical • Budget split

For 5QI 3 (PDB 50 ms) and 5QI 82 (PDB 10 ms), compute D_{AN} when the UPF is 600 km away (5 µs/km, plus 2 ms processing) and when it is local (0.5 ms). Which 5QI forces an edge UPF?

Exercise 4 • Level 2 – numerical • HARQ

A flow needs PER 10^{-6}. Compute the number of transmissions for $p = 0.1$ and $p = 0.001$, and the time needed with a HARQ round trip of 1 ms. What does this imply for the choice of MCS?

Exercise 5 • Level 3 – analytical • Admission control

A cell carries 5QI 1 voice (GBR 40 kbit/s per call at the radio, assumed to cost 0.5 % of cell capacity per call). Using Eq. (4) with voice as class 1, how many calls can be admitted so that non-GBR traffic at total load 0.9 sees a mean delay at most twice its FIFO value? Interpret.

Exercise 6 • Level 3 – analytical • End-to-end QoS

A cloud-gaming application needs 30 ms RTT and 50 Mbit/s downlink. Propose 5QI(s), UPF placement, slice and transport measures, and explain which nodes enforce which parameter.

Solutions

Solution to Exercise 1

One N3 tunnel (per PDU session) in all cases. DRBs: at least one (all three flows mapped to one DRB) and at most three (one per flow); the gNB decides.

Solution to Exercise 2

The UE derives uplink rules from downlink packets marked with the RQI, so no NAS signalling is needed per flow – valuable for many short-lived flows. It should not be used where the uplink treatment must differ from what the downlink packet implies, or for security-sensitive policy where the network must control exactly which uplink packets get privileged treatment (a reflected rule could be abused by crafted downlink traffic).

Solution to Exercise 3

Remote UPF: $600 \times 5\,\mu\text{s} = 3\,\text{ms} + 2\,\text{ms} = 5\,\text{ms}$. 5QI 3: $50 - 5 = 45\,\text{ms}$ (remote) or 49.5 ms (local); 5QI 82: $10 - 5 = 5\,\text{ms}$ (remote) or 9.5 ms (local). 5QI 82 loses half its access budget with the remote UPF – it needs an edge UPF; 5QI 3 does not.

Solution to Exercise 4

$p = 0.1$: $n = 6$, about 6 ms; $p = 0.001$: $n = 2$, about 2 ms. A conservative MCS (low BLER) costs spectral efficiency but saves retransmissions and delay – the characteristic URLLC trade-off of capacity for latency and reliability.

Solution to Exercise 5

Requirement: $T_2 \le 2T_{\text{FIFO}} \Leftrightarrow 1/(1-\rho_1) \le 2 \Leftrightarrow \rho_1 \le 0.5$: at most 100 calls (0.5 % each). In practice operators limit GBR load much lower (e.g. 20–30 %) to keep non-GBR delay close to FIFO and to keep margin for channel variations that change the cost per call.

Solution to Exercise 6

5QI: a GBR video-type value (e.g. 3 for real-time gaming characteristics) for the game stream, with GFBR set to the needed rate or a high-priority non-GBR 5QI with sufficient Session-AMBR; 5QI 5/default for control. UPF at a regional edge so that the core part of the RTT is a few milliseconds; possibly an eMBB/gaming slice with RAN resource shares in busy cells; transport mapping of the flow to a priority class. Enforcement: UPF (classification, MFBR, AMBR), gNB (GFBR, scheduling priority, PDB), UE (uplink rules); the PCF authorises it based on the gaming provider's AF request.

13 Further reading

The 5G QoS model and the standardised 5QI table are in TS 23.501 [5] (clause 5.7), policy and charging control in TS 23.503 [3], QoS procedures in TS 23.502 [4], SDAP and RAN aspects in TS 38.300 [1]. The LTE counterpart is TS 23.203 [2]. Kleinrock [7] derives priority queueing results; Dahlman et al. [6] discuss URLLC techniques in the NR radio.

References

[1] 3GPP, *NR; NR and NG-RAN overall description; stage-2*, TS 38.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38300.htm.

[2] 3GPP, *Policy and charging control architecture*, TS 23.203, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23203.htm.

[3] 3GPP, *Policy and charging control framework for the 5G system (5GS); stage 2*, TS 23.503, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23503.htm.

[4] 3GPP, *Procedures for the 5G system (5GS)*, TS 23.502, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23502.htm.

[5] 3GPP, *System architecture for the 5G system (5GS)*, TS 23.501, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23501.htm.

[6] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic Press, 2020.

[7] L. Kleinrock, *Queueing Systems, Volume 1: Theory*. Wiley, 1975.